

# User Lifecycle and Provisioning

**Series:** IAM | **Notebook:** 6 of 10 | **Created:** January 2026 | **Last Updated:** 02/27/2026

## Automating User Management at Scale

Manual user management doesn't scale. This notebook covers user lifecycle automation including SCIM provisioning, JIT access, service accounts, and token management.

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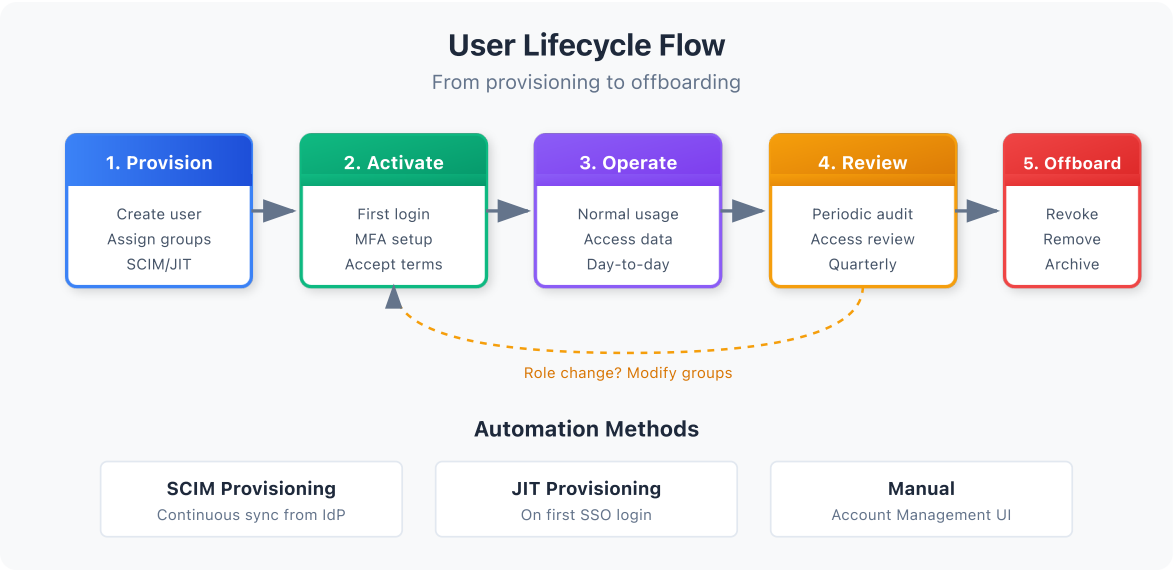
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## Prerequisites

| Requirement           | Details                                                  |
|-----------------------|----------------------------------------------------------|
| Dynatrace Environment | SaaS with Gen3 IAM enabled                               |
| Permissions           | <code>account-iam-admin</code> for user/token management |
| IdP                   | Enterprise IdP (Okta, Azure AD, etc.) for SCIM           |

# 1. User Lifecycle Overview

Users have a lifecycle from creation to deactivation. Effective management requires automation at each stage.



## Lifecycle Stages

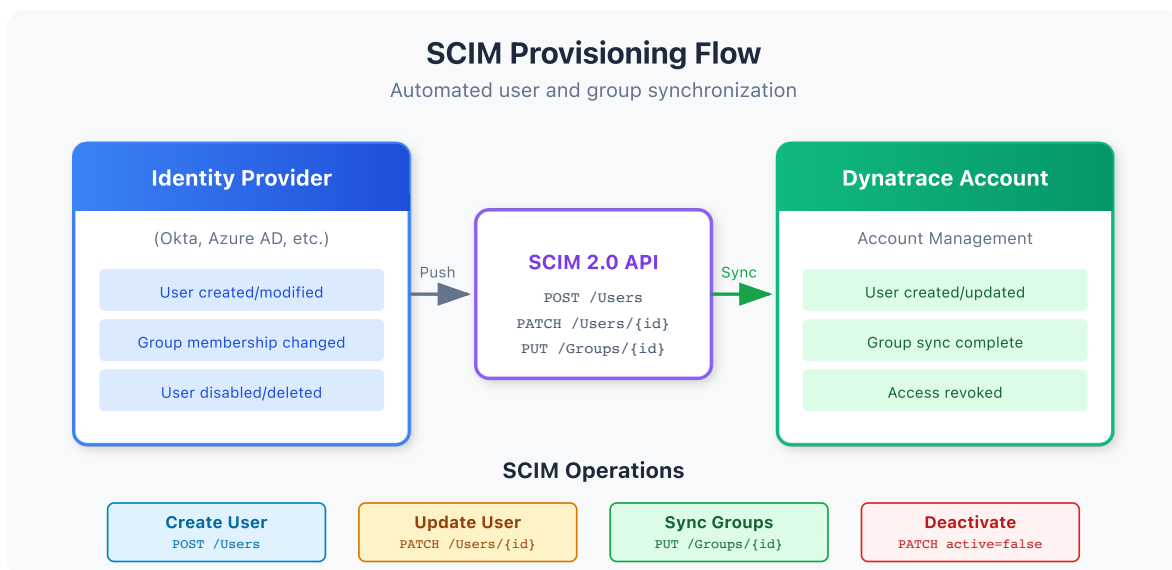
| Stage     | Description                 | Automation              |
|-----------|-----------------------------|-------------------------|
| Provision | Create user identity        | SCIM, JIT               |
| Assign    | Add to groups, grant access | SCIM group sync         |
| Activate  | First login, MFA setup      | SSO integration         |
| Operate   | Normal access and usage     | N/A                     |
| Review    | Periodic access review      | Audit queries           |
| Modify    | Change groups/access        | SCIM update             |
| Offboard  | Remove access               | SCIM delete, JIT expiry |

## Manual vs Automated

| Method | Pros                | Cons              | Scale              |
|--------|---------------------|-------------------|--------------------|
| Manual | Full control        | Slow, error-prone | < 50 users         |
| SCIM   | Automated sync      | Setup complexity  | 50 - 10,000+ users |
| JIT    | No pre-provisioning | Less control      | Any size           |

## 2. SCIM Provisioning

SCIM (System for Cross-domain Identity Management) automatically syncs users and groups from your IdP to Dynatrace.



## How SCIM Works

1. You configure SCIM in your IdP (Okta, Azure AD, etc.)
2. When users/groups change in IdP, SCIM syncs to Dynatrace
3. Users are automatically created, updated, or deactivated
4. Group memberships stay in sync

## Supported IdPs

| IdP                 | SCIM Support | Notes                       |
|---------------------|--------------|-----------------------------|
| Okta                | Full         | Native integration          |
| Azure AD (Entra ID) | Full         | Enterprise app provisioning |
| OneLogin            | Full         | Directory provisioning      |
| PingFederate        | Full         | Provisioning connector      |
| Google Workspace    | Partial      | Requires adapter            |

## SCIM Configuration Steps

### In Dynatrace:

1. Navigate to Account Management → Identity providers
2. Select your IdP configuration
3. Enable SCIM provisioning
4. Copy the SCIM endpoint URL and token

### In Your IdP (Example: Okta):

1. Open your Dynatrace SAML application
2. Go to Provisioning tab
3. Configure SCIM connection:
  - SCIM connector base URL
  - API token (from Dynatrace)
4. Enable provisioning features:
  - Create Users
  - Update User Attributes
  - Deactivate Users
  - Push Groups

## SCIM Attributes Synced

| Attribute  | Source               | Dynatrace Field  |
|------------|----------------------|------------------|
| userName   | IdP email            | Email (login)    |
| givenName  | IdP first name       | First name       |
| familyName | IdP last name        | Last name        |
| active     | IdP status           | Account status   |
| groups     | IdP group membership | Group membership |

## Best Practices

- **Test in sandbox** before production
- **Map groups explicitly** - don't auto-create
- **Use consistent naming** between IdP and Dynatrace
- **Monitor sync status** for failures

## 3. Just-in-Time Provisioning

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JIT provisioning creates users automatically on first login via SSO.

### How JIT Works

1. User authenticates via SAML SSO
2. If user doesn't exist, Dynatrace creates them
3. Group memberships assigned based on SAML assertion
4. User accesses Dynatrace immediately

## JIT vs SCIM

| Aspect       | JIT                     | SCIM           |
|--------------|-------------------------|----------------|
| User created | On first login          | Before login   |
| Deactivation | Manual or session-based | Automatic sync |
| Group sync   | At login only           | Continuous     |
| Setup        | Simple                  | More complex   |
| Control      | Less                    | More           |

## Configuring JIT

1. Navigate to Account Management → Identity providers
2. Select your SAML configuration
3. Enable "Auto-provision users on login"
4. Configure default group for new users
5. Map SAML groups to Dynatrace groups

## JIT Group Mapping

SAML assertion includes groups, which are mapped to Dynatrace groups:

| SAML Group          | Dynatrace Group     |
|---------------------|---------------------|
| IT-Dynatrace-Admins | dt-platform-admins  |
| App-Checkout-Team   | dt-checkout-editors |
| All-Employees       | dt-all-viewers      |

## JIT Considerations

| Pro                        | Con                            |
|----------------------------|--------------------------------|
| Simple setup               | User created only after login  |
| No pre-provisioning needed | Group sync only at login       |
| Self-service access        | Harder to audit who has access |

## 4. User Onboarding Workflows

### Standard Onboarding Process

| Step | Action                       | Owner        |
|------|------------------------------|--------------|
| 1    | Add user to IdP group        | HR/IT        |
| 2    | SCIM syncs user to Dynatrace | Automatic    |
| 3    | User receives welcome email  | IT           |
| 4    | User logs in via SSO         | User         |
| 5    | User completes MFA setup     | User         |
| 6    | Access verified              | User/Manager |

### New Employee Checklist

- ❑ Employee added to corporate directory
- ❑ Employee assigned to appropriate IdP groups
- ❑ SCIM sync verified (user appears in Dynatrace)
- ❑ Employee logs in successfully
- ❑ Employee completes MFA enrollment
- ❑ Employee confirms access to required data

### Role Change Process

When users change roles:

1. Manager requests access change
2. IAM admin approves
3. Update IdP group membership
4. SCIM syncs changes
5. User has new access on next login

## Temporary Access

For contractors or short-term access:

| Method                 | Duration       | Automation        |
|------------------------|----------------|-------------------|
| IdP time-limited group | Days to months | IdP policy        |
| Manual group removal   | Any            | Calendar reminder |
| OAuth client expiry    | Hours to days  | Token config      |

## 5. User Offboarding

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Proper offboarding is critical for security.

### Offboarding Triggers

| Trigger      | Response           | Timing    |
|--------------|--------------------|-----------|
| Termination  | Full revocation    | Immediate |
| Resignation  | Full revocation    | Last day  |
| Role change  | Partial revocation | At change |
| Contract end | Full revocation    | End date  |

### Automated Offboarding (SCIM)

1. User disabled/removed in IdP
2. SCIM syncs deactivation to Dynatrace
3. User cannot login



#### 4. Existing sessions invalidated

### Manual Offboarding Checklist

- ☐ Remove from all Dynatrace groups
- ☐ Revoke any API tokens created by user
- ☐ Deactivate OAuth clients owned by user
- ☐ Transfer ownership of dashboards/notebooks
- ☐ Review audit log for recent activity
- ☐ Document offboarding completion

### Token Revocation on Offboarding

Critical: User-created tokens remain active until explicitly revoked.

#### Query to find user's tokens:

```
Account Management → API tokens  
Filter by: Created by [user email]  
Action: Revoke all tokens
```

### Verification

After offboarding, verify:

- ☐ User cannot login via SSO
- ☐ User not in any groups
- ☐ User's tokens are revoked
- ☐ No active OAuth clients

## 6. Service Accounts and OAuth Clients

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Non-human identities require special handling.

# Service Accounts vs OAuth Clients

| Type            | Use Case            | Authentication |
|-----------------|---------------------|----------------|
| Service Account | Automation, scripts | API token      |
| OAuth Client    | Apps, integrations  | OAuth 2.0      |

## Creating OAuth Clients

- 1. Account Management → OAuth clients
- 2. Create new client
- 3. Configure:
  - Name and description
  - Grant type (client credentials)
  - Scopes (permissions)
  - Token lifetime
- 4. Save client ID and secret securely

## OAuth Client Best Practices

| Practice                   | Rationale                  |
|----------------------------|----------------------------|
| One client per application | Isolation, easy revocation |
| Minimum scopes             | Least privilege            |
| Short token lifetime       | Reduce exposure            |
| Rotate secrets regularly   | Security hygiene           |
| Document ownership         | Accountability             |

## OAuth Scopes

Common scopes for automation:

| Scope                                | Grants          |
|--------------------------------------|-----------------|
| <code>storage:logs:read</code>       | Read logs       |
| <code>storage:metrics:read</code>    | Read metrics    |
| <code>settings:objects:read</code>   | Read settings   |
| <code>settings:objects:write</code>  | Modify settings |
| <code>document:documents:read</code> | Read dashboards |

## Service Account Lifecycle

| Event              | Action                          |
|--------------------|---------------------------------|
| Create             | Document owner, purpose, expiry |
| Review (quarterly) | Verify still needed             |
| Owner change       | Update documentation            |
| Decommission       | Revoke client, archive docs     |

## 7. API Token Management

API tokens provide access for scripts, tools, and legacy integrations.

### Token Types

| Type            | Scope              | Use Case                |
|-----------------|--------------------|-------------------------|
| <b>Personal</b> | User's permissions | User automation         |
| <b>Service</b>  | Specific scopes    | Application integration |

## Token Scopes Reference

| Scope                       | Permission       |
|-----------------------------|------------------|
| <code>logs.read</code>      | Read log data    |
| <code>metrics.read</code>   | Read metrics     |
| <code>entities.read</code>  | Read entity data |
| <code>settings.read</code>  | Read settings    |
| <code>settings.write</code> | Modify settings  |
| <code>problems.read</code>  | Read problems    |
| <code>events.ingest</code>  | Send events      |

## Token Best Practices

| Practice          | Implementation                            |
|-------------------|-------------------------------------------|
| Minimum scopes    | Only grant what's needed                  |
| Descriptive names | <code>cicd-pipeline-metrics-reader</code> |
| Expiration dates  | Set reasonable expiry                     |
| Rotation schedule | Rotate every 90 days                      |
| Secure storage    | Use secrets manager                       |
| No sharing        | One token per use case                    |

## Token Rotation Process

1. Create new token with same scopes
2. Update applications with new token
3. Verify applications work
4. Revoke old token
5. Document rotation

## Monitoring Token Usage

Use audit logs to track token activity.

```
// Track API token usage patterns
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "token")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Find token creation events
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "token") and matchesPhrase(content,
"created")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Monitor OAuth client activity
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "oauth") or matchesPhrase(content, "client")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

## Next Steps

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With user lifecycle management in place, proceed to compliance:

### Recommended Path

1. **IAM-07: Audit Logging and Compliance** - Monitor and report on access
2. **IAM-08: Multi-Environment IAM** - Scale across environments
3. **IAM-09: Troubleshooting Access Issues** - Debug problems

## User Lifecycle Checklist

Before moving on, ensure you have:

- ☐ Chosen SCIM, JIT, or manual provisioning
  - ☐ Configured IdP integration for provisioning
  - ☐ Documented onboarding process
  - ☐ Documented offboarding process
  - ☐ Established token management policy
  - ☐ Created service account inventory
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## Summary

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In this notebook, you learned:

- User lifecycle stages and automation options
  - SCIM provisioning configuration
  - Just-in-time provisioning via SSO
  - User onboarding and offboarding workflows
  - Service account and OAuth client management
  - API token best practices and rotation
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## References

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- [SCIM Provisioning](#)
  - [OAuth Clients](#)
  - [API Tokens](#)
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